

MATTHEW DUNN

matthew.thomas.dunn@gmail.com ◇ Google Scholar

EXPERIENCE

CTO

Research Therapeutics

July 2026 - Present

Boston, MA

Lead technology and scientific computing strategy for an early-stage therapeutics company developing lipid nanoparticle-based medicines. Support company formation and fundraising while building machine learning systems for therapeutic development, including protein language models for lipid nanoparticle design and optimization. Develop AI-assisted research platforms for large-scale, unstructured literature analysis, evidence synthesis, and scientific diligence, and conduct computational and background research across formulation, targeting, immunology, translational strategy, and other company R&D efforts.

Co-Founder & CEO

Daedaline

October 2023 - July 2026

Boston, MA

Raised \$2M in seed financing, recruited the initial team, and built Daedaline from concept into a production platform for AI-assisted real-asset due diligence. Architected the system and personally implemented core components, including hybrid SOTA retrieval, distributed inference, and agentic query-routing pipelines for high-throughput diligence workflows. Established external partnerships and negotiated customer and partner contracts, carrying the company across fundraising, product, engineering, and go-to-market execution.

Research Director - Member of the Technical Staff

Cohere

July 2022 - October 2023

Boston, MA

At Cohere, I established the company's annotation system strategy to support scalable, high-quality training data pipelines. I defined the forward-deployed engineer strategy to improve customer integration and accelerate adoption of language models in production settings. I also designed and prototyped post-training methods for model adaptation on Google TPUs, focusing on aligning large language models with complex commercial applications through retrieval integration and chain-of-thought reasoning.

VP of Applied Machine Learning

LivePerson

Sept 2021 - April 2022

Boston, MA

Oversaw company-wide deployment of deep learning methods across all product lines. Expanded and led a 20-person applied research team, primarily PhDs, that delivered LivePerson's flagship AI product and advanced models for intent classification, dialogue management, entity recognition, sentiment analysis, contact center optimization, and forecast modeling, supporting Tier 1 services in eight languages at global scale.

Director of Applied Machine Learning

LivePerson

Sept 2018 - Sept 2021

Seattle, WA

I recruited a team of applied NLP researchers and supported them in developing AI products from initial definition through GA release in collaboration with partner engineering teams. I was responsible for my team's delivery of machine learning research that supports flagship AI products across the LivePerson Conversational Cloud offering. We trained BERT and related transformer variants on one of the largest (human-to-human) conversational data sets. My team supports numerous NLP models that underpin Tier 1 services.

Data Scientist - NLP*LivePerson*

Apr 2017 - Sept 2018

New York, NY

I worked with the CEO, Rob LoCascio, to hire and train an applied machine learning research team to focus on developing AI products for strategic initiatives. We developed products using unsupervised, semi-supervised, and supervised NLP techniques to extract commercially valuable insights from (human-to-human) conversational data sets.

Graduate Student Researcher*NYU Center for Data Science*

Aug 2016 - May 2017

New York, NY

Member of the CILVR group at NYU where I worked on the development of the SearchQA data set and associated models to solve open domain Question and Answer tasks. Ported NLP models (LSTM, GRU, etc.) from Theano to Pytorch contributing to Pytorch public release.

Data Scientist*Wylan Energy*

Summer 2016

New York, NY

Developed quantitative simulation and forecasting models to support participation in U.S. wholesale electricity capacity auctions across PJM, ISO-NE, and MISO. Built statistical frameworks to estimate future capacity value derived from distributed energy efficiency resources, incorporating pricing uncertainty, demand variability, and regulatory constraints. Designed probabilistic bidding and portfolio performance models using Monte Carlo and scenario analysis techniques, and translated complex market design rules into practical decision tools for resource valuation and strategy optimization.

CEO*MD Consulting*

2014 - 2015

New York, NY

Built a laboratory information management system for a biotech to improve research and development efforts for high throughput genetic sequencing diagnostic tests for colon cancer. Managed a team of contractors in addition to doing hands on engineering.

Software Engineer 2012 - 2014 – Exponent Partners (B-Corp)**Software Engineer** 2010 - 2012 – Groundswell (501.c.3)**RESEARCH - GOOGLE SCHOLAR**

Actionable Conversational Quality Indicators for Improving Task-Oriented Dialog Systems
Natural Language Engineering*PDF*

2022

Domain-specific Distillation Models for Conversational Commerce ACL Workshop ECNLP*Accepted Papers*

2022

SearchQA: A New QA Dataset Augmented with Context from a Search Engine*PDF*

2017

Early Predictability of Asylum Court Decision

ACM Conference on AI and the Law

PDF

2017

Optimizing Targeted Sequencing of Somatic Mutations

2016

CONFERENCE REVIEWER

NeurIPS	2025
ACL	2023, 2025, 2026
NAACL	2019, 2021, 2022, 2024
EMNLP	2025, 2026
EACL	2026
COLING	2020
IJCAI-ECAI	2026

PATENTS

Domain Adaptation of AI NLP Encoders with Knowledge Distillation	Patent No.: US-20220318502-A1
Automatic Bot Creation Based on Scripts	Patent No.: US 10,853,568 B2
Intent Driven Contact Center	Patent No.: US 11,050,884 B2
Systems and Methods for Intent Analysis	Patent No.: US 10,902,199 B2

EDUCATION

M.S. Data Science, NYU Courant Institute - CILVR Lab - ML^2 Group
BA Environmental Economics, University of Redlands - Minor: English Literature

VOLUNTEERING ACTIVITIES

Advisor 2019-Present
NYU Center for Data Science

I work with NYU Center for Data Science capstone groups to develop and execute their research proposals over the course of the semester.